Pyruvate kinase activators in hereditary haemolytic anaemias: current evidence and clinical potential

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Hereditary haemolytic anaemias represent the most prevalent group of genetic disorders worldwide and have a substantial impact on global health. Current treatments are few and primarily supportive. Recent studies suggest a crucial and overlapping role of metabolic impairment of red blood cells in these diseases, extending beyond the primary genetic defect. Pyruvate kinase activators enhance glycolysis, thereby targeting this shared metabolic impairment by increasing ATP production and improving cellular homeostasis. The first pyruvate kinase activator has been approved for the treatment of pyruvate kinase deficiency. Clinical trials evaluating pyruvate kinase activators in other haemolytic disorders, including thalassaemia, sickle cell disease, and red blood cell membrane disorders have provided evidence of clinical efficacy by ameliorating haemolytic anaemia and improving other disease-related outcomes, while maintaining a generally favourable safety profile. Ongoing preclinical and translational research continues to provide further insights into other potential indications for pyruvate kinase activators.

Lancet 2026; 407: 1383–96

Published Online
March 12, 2026
https://doi.org/10.1016/S0140-6736(26)00150-9

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Introduction

Hereditary haemolytic anaemias comprise a wide range of disorders, including haemoglobinopathies, red blood cell membrane disorders, and enzyme deficiencies. These disorders are characterised by haemolysis, splenomegaly, and secondary haemochromatosis, with phenotypes ranging from mild anaemia to life-threatening, transfusion-dependent haemolysis.[1,2]

Current treatment strategies are primarily supportive, including blood transfusions, iron chelation, and splenectomy. Curative options, including allogenic stem cell transplantation and emerging gene therapies, carry substantial risks and remain inaccessible to most.[3] Consequently, a substantial unmet therapeutic need remains, especially for symptomatic patients with mild-to-moderate disease burden and those not eligible or comfortable with the current risks of curative interventions.

Pyruvate kinase (PK) activators represent a new targeted therapeutic approach for hereditary haemolytic anaemias, addressing a fundamental metabolic vulnerability in red blood cells. PK activators enhance glycolytic flux and ATP generation within red blood cells, counteracting the depletion that contributes to haemolysis. Advances in the understanding of the role of cellular metabolism in other hereditary haemolytic anaemias have shown an overlapping role of glycolytic impairment, not limited to enzyme deficiencies.[4–6] These insights have led to the exploration of PK activators across PK deficiency and other hereditary haemolytic anaemias, including thalassaemia, sickle cell disease, and red blood cell membrane disorders.[7–9]

This Therapeutics paper provides a comprehensive overview of red blood cell metabolism and its essential role in supporting red blood cell survival. The therapeutic rationale and the mechanisms of action of PK activators across hereditary haemolytic anaemias are explained in addition to the current evidence on safety and efficacy outcomes, future directions, and clinical potential of this emerging therapeutic class.

Red blood cell metabolism: the basis for cellular homeostasis and survival

Glycolytic dependency of red blood cells and the key role of PK

Mature, non-mutated red blood cells do not have mitochondria and are therefore incapable of generating ATP through mitochondrial cellular respiration, relying exclusively on anaerobic glycolysis.[10] PK catalyses the final and irreversible step in glycolysis, yielding one ATP molecule. As glycolytic intermediates are cut in half by aldolase before reaching PK, the net yield is two molecules of ATP per glucose molecule.[11] The total net production of glycolysis is therefore dependent on PK. The other product of PK, pyruvate, serves as an important substrate for several subsequent metabolic pathways in nucleated cells.[12]

The human body contains four different PK isoenzymes, encoded by two distinct genes. *PKM* encodes PKM1 and PKM2, found in skeletal muscle, the heart, the brain, and in fetal and several mature tissues, including proliferating cells. *PKLR* encodes PKL, found primarily in the liver and PKR, confined to the red blood cell.[12,13] The PKR isoenzyme is a tetrameric enzyme, with two major conformational states: the T-state (tense) and

Search strategy and selection criteria

We searched the PubMed database for relevant publications from its inception to April 23, 2025, using search terms "pyruvate kinase activator", "PK activator", "mitapivat", "AG-348", "etavopivat", "FT-4202", "tebapivat", and "AG-946". The reference lists of the identified relevant publications were reviewed to identify additional publications of interest. Relevant clinical trials were identified in both the US National Library of Medicine clinical trials database, the EU Clinical Trials Register, and the WHO International Clinical Trials Registry Platform using the search terms above. Selection of relevant publications prioritised those published within the past 3 years, but older work was included where relevant.

the active R-state (relaxed). The enzyme exists in an equilibrium between the two states, which transitions to the active R-state upon binding of endogenous allosteric activators, such as fructose-1,6-bisphosphate, at its regulatory site. Because the R-state has an increased affinity for its substrate, phosphoenolpyruvate, PK activity, and glycolytic flux become enhanced, allowing more efficient catalysis.[14]

Energy-driven mechanisms in red blood cell maintenance and survival

Adequate glycolytic functioning is essential for ATP generation, and glycolytic impairment causes haemolysis, as seen in PK deficiency.[10] ATP supports several important energy-driven mechanisms that are essential to cellular functioning and survival. These mechanisms include regulation and maintenance of oxidative and metabolic homeostasis; support of red blood cell deformability, measured by membrane integrity, flexibility, and cellular hydration; preservation of membrane asymmetry; and effective erythropoiesis. Impaired glycolysis leads to dysfunction of these mechanisms, consequently resulting in anaemia. Although mechanisms in the following section and figure 1 are presented individually, they are highly interrelated.

Oxidative and metabolic homeostasis

ATP is required to convert glucose into glucose-6-phosphate, which fuels the hexose monophosphate pathway (HMP). This pathway generates nicotinamide adenine dinucleotide phosphate (NADPH), used to regenerate reduced glutathione. Glutathione is a key antioxidant that protects haemoglobin, glycolytic enzymes, and membrane proteins from oxidative damage.[15,16] Severe ATP depletion can impair glucose phosphorylation, reducing HMP substrate availability and potentially limiting NADPH production.[15] Consequently, glutathione regeneration might be restricted, resulting in cumulative oxidative stress.[16,17] Older functional studies reported conflicting findings on whether HMP shunt capacity is diminished in small case series of glycolytic enzyme deficiencies with severe ATP depletion.[18,19] In PK deficiency, one such study found reduced HMP shunt activity following oxidative stress, but this was based on only five patients.[18] Metabolomics on peripheral blood, however, also suggest reduced HMP activity in PK deficiency compared with healthy controls.[20] In disorders such as sickle cell disease and hereditary spherocytosis, oxidative stress contributes to haemolysis by inhibiting phosphatases and activating kinases, causing excessive tyrosine phosphorylation of band 3, which destabilises the membrane–cytoskeletal network and promotes red blood cell destruction.[21,22]

Cellular deformability

Red blood cell deformability is measured by membrane flexibility, cytoskeletal integrity, and cellular hydration, all primarily regulated by ATP-dependent processes.[23] Decreased deformability causes splenic sequestration and extravascular haemolysis, and might also contribute to intravascular haemolysis in sickle cell disease.[23] ATP depletion in red blood cells prevents ATP-dependent protein 4.1R (P4.1) phosphorylation, strengthening membrane–spectrin interactions, thereby reducing membrane flexibility.[24] Moreover, ATP depletion indirectly leads to oxidative stress-related hyperphosphorylation of band 3.[21] This substantially alters the membrane's structure and function, also affecting membrane-bound glycolytic enzymes and ion-transporters.[22,25] Lastly, insufficient ATP levels impair plasma membrane Ca^{2+}-ATPase functioning, causing intracellular Ca^{2+} concentrations to rise.[26] This activates the Ca^{2+}-sensitive K^+ (Gardos) channels, leading to K^+ and water efflux, resulting in cellular dehydration. This mechanism underlies hereditary xerocytosis but also occurs in PK deficiency and sickle cell disease.[23] High Ca^{2+} concentrations also activate proteolytic enzymes such as calpain, which disrupt membrane–protein interactions and induce membrane vesiculation.[27] Collectively, these mechanisms reduce cellular deformability, ultimately promoting haemolysis.

Membrane asymmetry

ATP is required for maintaining membrane asymmetry by driving energy-dependent transport proteins.[28] The confinement of phosphatidylserine to the inner layer is important as phosphatidylserine exposure is prothrombotic and induces phagocytosis by splenic macrophages.[23] ATP depletion leads to increased intracellular Ca^{2+}, which enhances scramblase activity and inhibits flippase activity, resulting in increased phosphatidylserine exposure.[23,29] This mechanism thus contributes to haemolysis, as has been observed in sickle cell disease and thalassaemia.[26,30]

Erythropoiesis

Erythroid maturation is energy demanding, with maturing red blood cells progressively shifting from aerobic mechanisms towards anaerobic glycolysis, making glycolytic functioning especially important during terminal erythroid stages.[31] Glycolytic impairment might contribute to ineffective erythropoiesis via several mechanisms. In PK deficiency, reduced pyruvate generation forces red blood cell precursors to use alternative anaplerotic carbon entry (primarily α-ketoglutarate derived from glutamine) as fuel for the tricarboxylic acid cycle. Consuming other carbons in early maturation stages presumably reduces antioxidant potential at later stages, contributing to defective mitophagy and phagocytosis of precursors.[32,33] In thalassaemia, ineffective erythropoiesis arises from excess unpaired α-globin or β-globin chains within erythroblasts, causing chronic oxidation and apoptosis of red blood cell precursors.[34,35] Reduced ATP generation could aggravate this process, as clearance of these chains relies on ATP-dependent proteolytic mechanisms.[8] Lastly, phosphatidylserine exposure in erythroid precursors, triggered by low-energy states, also contributes to ineffective erythropoiesis by promoting their phagocytic removal.[30]

Biological rationale for PK activators across hereditary haemolytic anaemias

ATP is essential for red cell homeostasis, supporting metabolism, redox balance, membrane integrity, hydration, and deformability, as well as erythropoiesis. Anaemias with impairment of any of these mechanisms might benefit from treatment with PK activators. Red blood cells in various hereditary haemolytic anaemias

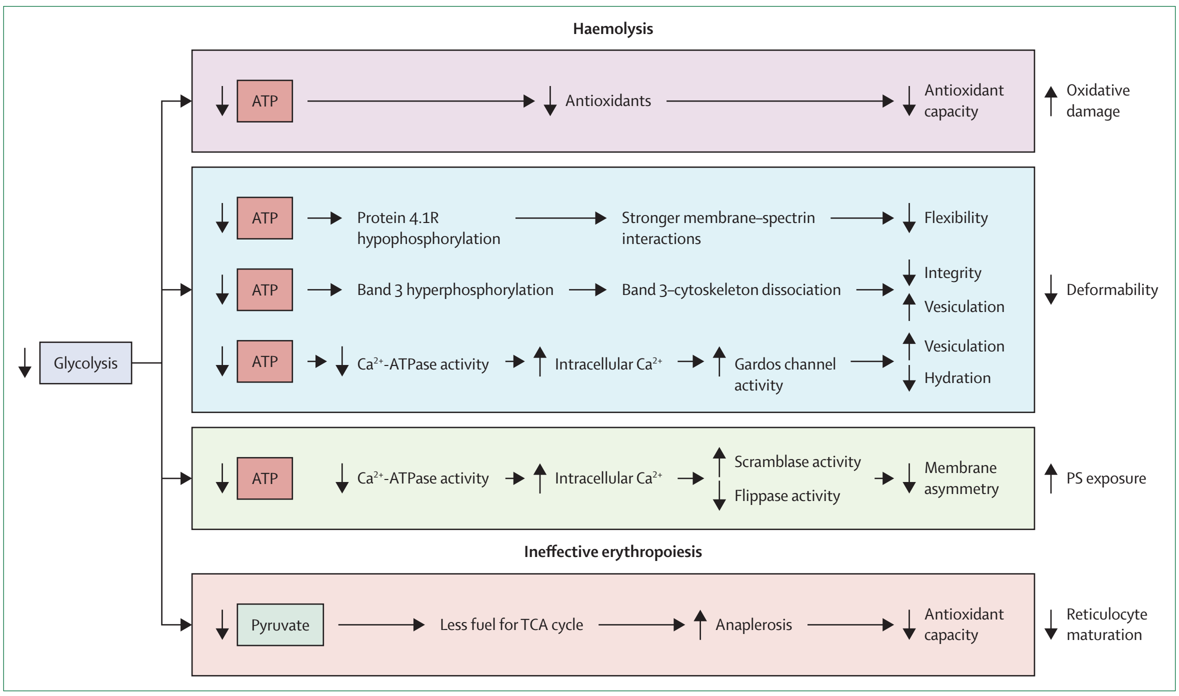

***Figure 1:* Pathways through which glycolytic dysfunction causes anaemia**
Impaired glycolytic function leads to anaemia through several mechanisms affecting red blood cell function, homeostasis, production, and survival. PS=phosphatidylserine. TCA=tricarboxylic acid. Figure created using biorender.com.

	Mechanism of action	Diseases investigated in clinical trials	Dosages in phase 2/3 clinical trials, mg	Median range t_{max}, h* (min, max)	Mean range $t_{½}$, h* (±SD)	Bioavailability† (%)	Metabolism	Elimination
Mitapivat (AG-348)[13,40,41]	Allosteric activator of PKR, PKM2, and PKL	PKD, SCD, thalassaemia, RBC membrane disorders, and CDA II	5–200 BID	0·77 (0·48, 2·00)–1·01 (0·50, 3·00)‡	17·8 (3·3)–20·4 (3·6)‡	72·7	CYP3A4, CYP3A5, CYP1A2, CYP2C8, CYP2C9	Primarily via hepatic metabolism (1·1–2·4% renal clearance)
Etavopivat (FT-4202)[38,42]	Selective allosteric PKR activator	SCD and thalassaemia	200–400 QD	0·50 (0·5, 3·0)–1·5 (0·5, 4·0)	10·6 (2·4)–13·8 (3·9)	..	CYP3A4, CYP3A5, CYP2C9	Primarily via hepatic metabolism (0·8–1·7% renal clearance)
Tebapivat (AG-946)[39]	Allosteric activator of PKR and PKM2	SCD and MDS	2·5–20 QD	..	..	..	..	..

t_{max}=time to maximum concentration. $t_{½}$=estimated terminal elimination half-life. PKR=pyruvate kinase R. PKD=pyruvate kinase deficiency. SCD=sickle cell disease. RBC=red blood cell. CDA II=congenital dyserythropoietic anaemia type 2. BID=twice daily. QD=once daily. MDS=myelodysplastic syndrome. *Results displayed are based on single dosage of either 30 mg, 120 mg, or 360 mg mitapivat or 200 mg, 400 mg, 700 mg, or 1000 mg etavopivat. †After administration of a single dose of 120 mg mitapivat. ‡Accuracy and reliability at higher dose levels (700–2500 mg) were influenced by duration of sampling and were therefore not included.

***Table 1:* Pharmacodynamic and pharmacokinetic properties of current pyruvate kinase activators**

exist in a state of chronic metabolic stress with relative shortage of ATP due to an increased ATP demand to fuel ATP-dependent compensatory mechanisms.[12,36] However, glycolytic functioning and ATP generation are often insufficient or even impaired, promoting anaemia through several mechanisms (figure 1).[4,5] By increasing glycolytic flux, ATP availability, and cell deformability, PK activators could improve anaemia and haemolysis across a range of hereditary haemolytic anaemias.

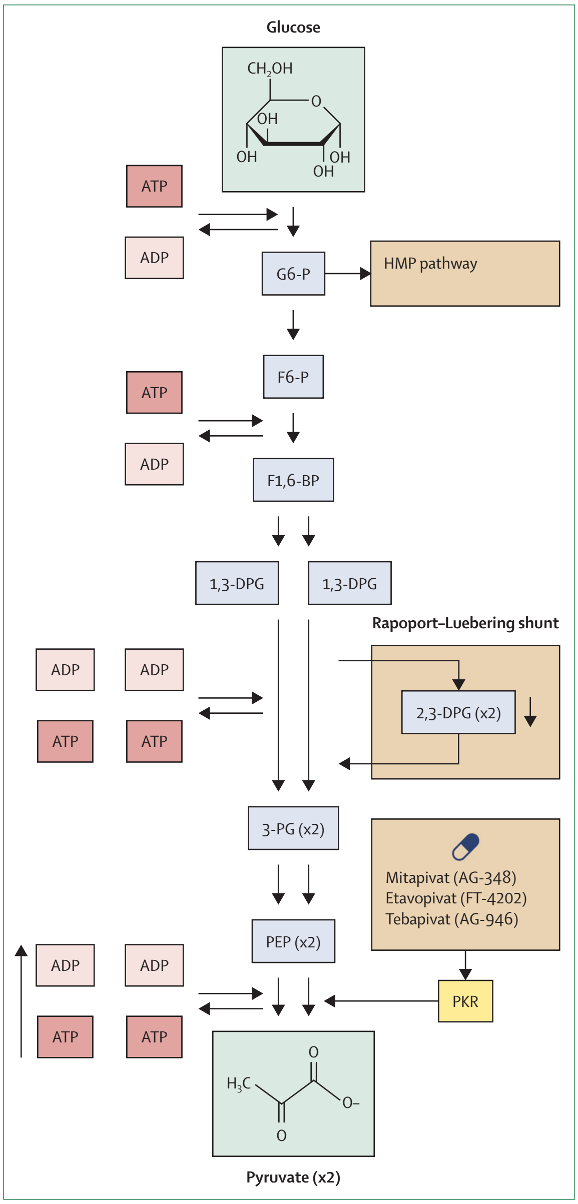

Figure 2: **Effects of PK activation on glycolytic flux**

A simplified overview of the glycolytic pathway, highlighting the effects on ATP and 2,3-DPG following PK activation. G6-P=glucose-6-phosphate. HMP=hexose monophosphate. F6-P=fructose-6-phosphate. F1,6-BP=fructose-1,6-biphosphate. 1,3-DPG=1,3-diphosphoglycerate. 2,3-DPG=2,3-diphosphoglycerate. 3-PG=3-phosphoglycerate. PEP=phosphoenolpyruvate. PK=pyruvate kinase. PKR=pyruvate kinase R. Figure created using biorender.com.

Mechanisms of action

PK activators enhance glycolytic flux, increasing ATP levels and decreasing 2,3-DPG levels

PK activators enhance PK activity by acting as an allosteric activator. The therapeutic agent was initially developed to target the PKM2 paradox in the Warburg effect through activation of the PKM2 isoform.[37] However, shortly thereafter, mitapivat (formerly AG-348), a small allosteric non-specific PK activator, was developed to target the mutant PK enzymes underlying the pathophysiology of PK deficiency.[13] Binding of mitapivat to a pocket at the dimer–dimer interface of the PK enzyme, distinct from the FBP binding site, leads to conformational changes that increase phosphoenolpyruvate binding affinity. Etavopivat (formerly FT-4202) is a potent and selective PKR activator. Etavopivat, similar to mitapivat, stabilises PKR, but has a higher selectivity for PKR.[38] Tebapivat (formerly AG-946) is a second-generation PK activator optimised with increased potency, longer residence time, and improved pharmacokinetics and pharmacodynamics, including longer half-life time (table 1).[39]

In the first preclinical study, mitapivat improved PKR activity across various mutant PK enzymes encompassing a range of PK deficiency genotypes. Ex vivo mitapivat treatment of red blood cells from both healthy controls and patients with PK deficiency resulted in dose-dependent increases in PKR activity and thermostability, accompanied by increases in ATP and decreases in 2,3-diphosphoglycerate (2,3-DPG) levels (figure 2).[13] Red blood cells from patients with PK deficiency of specific genotypes, for example deletions and nonsense variants, exhibited little activation of PK upon treatment. This was most likely due to severely reduced PK protein levels, thereby leaving little PK enzyme available for activation, a finding confirmed in clinical trials.[13,43] The observation that mitapivat also enhanced PK activity of the wild-type PKR enzyme suggested that patients with other red blood cell diseases characterised by relative ATP shortage, such as thalassaemia and sickle cell disease, could also benefit from PK activators. This observation has been the main rationale for investigating PK activators in other red blood cell diseases.[13]

In preclinical β-thalassaemia studies, including a murine model and an ex vivo human red blood cell model, mitapivat increased PK activity and ATP levels and ameliorated ineffective erythropoiesis, iron overload, and anaemia.[5,44] In preclinical sickle cell disease studies using red blood cells from mice, non-human primates, and humans, PK activators increased haemoglobin–oxygen affinity, reduced sickling, improved deformability, and decreased haemolysis.[45–47] These findings support a dual mechanism of action in sickle cell disease: increased ATP levels, leading to improved red blood cell homeostasis, enhanced deformability, and reduced haemolysis; and decreased 2,3-DPG levels with increased haemoglobin–oxygen affinity, reflected by reduced p50 levels, and reduced red blood cell sickling. This

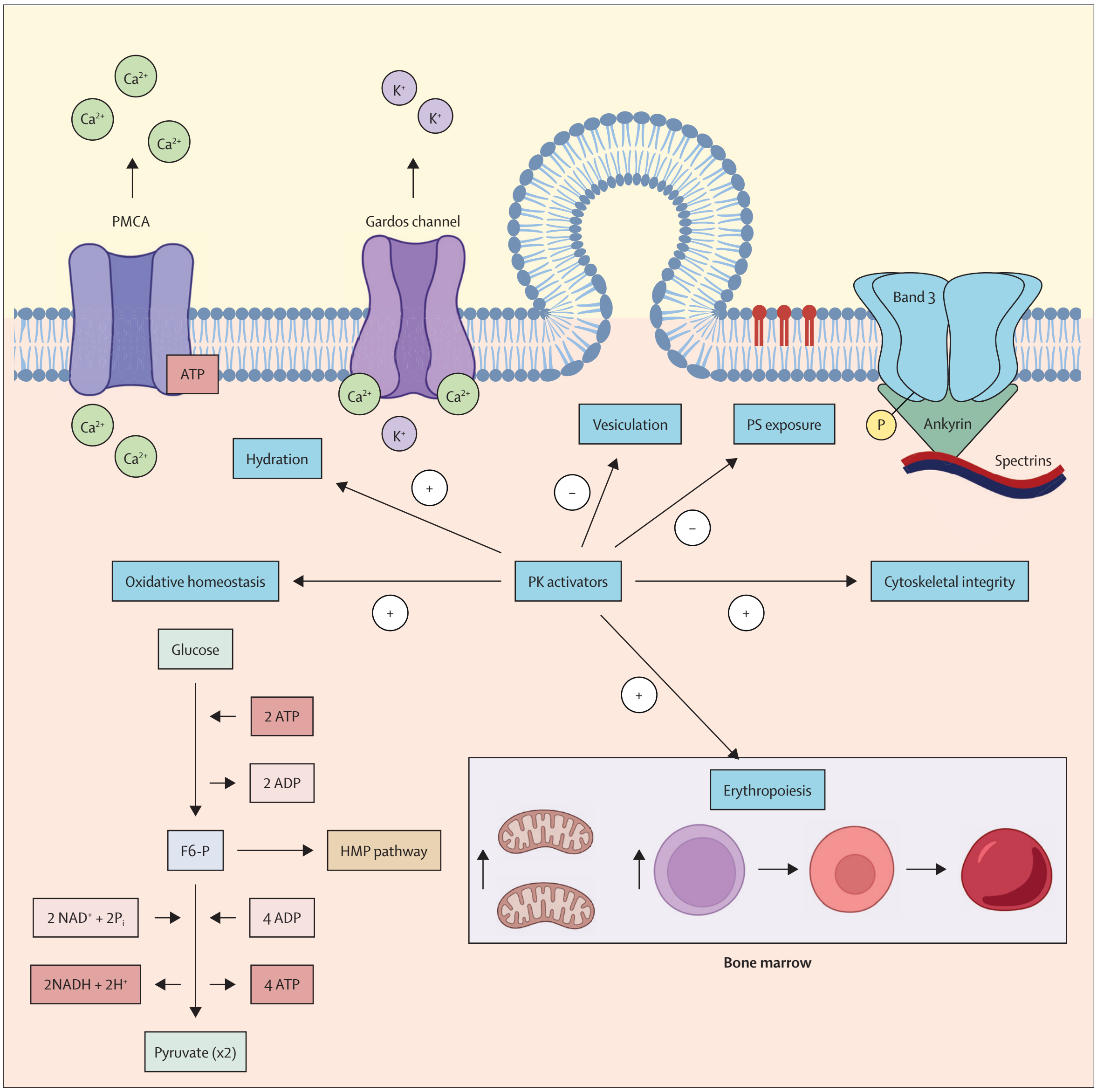

Figure 3: **Effect of PK activators on cellular mechanisms that influence cell survival**
Intracellular and intramedullary pathways improved by PK activation across hereditary haemolytic anaemias. Mechanisms described from left to right. PK activators enhance glycolytic flux, thereby increasing ATP production, decreasing glycolytic intermediates, including 2,3-diphosphoglycerate, and restoring the metabolic and oxidative homeostasis. Consequently, less oxidative stress-related protein denaturation and band 3 phosphorylation occur. Increased ATP availability following PK activation might lead to enhanced PMCA-mediated Ca^{2+} efflux. As a result, decreased intracellular Ca^{2+} concentrations prevent Gardos channel activation, thereby reducing K^+ efflux and cellular dehydration. Moreover, decreased intracellular Ca^{2+} concentrations prevent membrane vesiculation, and enable the maintenance of membrane asymmetry, including less PS exposure. Increased ATP generation resulting from PK activation boosts protein phosphatase activity, thereby preventing the phosphorylation of band 3. This effect leads to enhanced cytoskeleton–membrane interactions, preventing membrane vesiculation and improving membrane integrity and flexibility. During erythropoiesis, PK activators increase glycolytic flux in the erythroblasts, enhancing the oxidative stress response and promoting erythroid maturation. Additionally, PK activators enhance mitochondrial biogenesis and promote mitochondrial clearance, thereby ameliorating mitochondrial dysfunction. Both mechanisms contribute to improved erythropoiesis. F6-P=fructose-6-phosphate. HMP=hexose monophosphate. P=phosphate. PK=pyruvate kinase. PMCA=plasma membrane Ca^{2+}-ATPase. PS=phosphatidylserine. Figure created using biorender.com.

mechanism is likely driven by the mitigation of sickle haemoglobin (HbS) polymerisation.[48] In a sickle cell disease study, red blood cells showed reduced microvascular occlusion under both normoxic and hypoxic conditions following PK activation, reflecting improved deformability and reduced sickling through improved haemoglobin–oxygen affinity, limiting HbS polymerisation.[49] In a preclinical hereditary spherocytosis mouse model, mitapivat ameliorated anaemia and was non-inferior to splenectomy.[50]

Subsequent healthy volunteer phase 1 clinical trials of PK activators have shown findings consistent with preclinical studies, showing dose-dependent increases in ATP levels, haemoglobin–oxygen affinity, and decreases in 2,3-DPG concentration.[40,42]

PK activators drive several cellular pathways that enhance red blood cell survival

See Online for appendix

Figure 3 and appendix (p 1) present the effects of PK activation on cellular mechanisms. PK activators improve the metabolic and oxidative homeostasis of the red blood cell. In β-thalassaemic erythroblasts and a hereditary spherocytosis mouse model, mitapivat reprogrammed the red blood cell metabolome and ameliorated the intracellular redox milieu, supported by the downregulation of hypoxia-regulated genes and enzymes.[50,51] Similar findings were observed following tebapivat exposure to hereditary spherocytosis affected red blood cells ex vivo.[52] Although steady state metabolomics have numerous limitations, these results suggest that PK activation enhances the antioxidant status of red blood cells. Other observations underlining the

	Design	Dosage	Population, n	Treatment period	Safety outcomes*	Key efficacy outcomes
Mitapivat						
Yang et al[40] (NCT0210810)PK	Phase 1, randomised, double-blind, placebo-controlled, single ascending dose	30–2500 mg, single dose	Healthy volunteers, 36	Single dose	Headache (16·7%); nausea (13·9%); AE ≥grade 3 (0%)	ATP ↑; 2,3-DPG decrease
Yang et al[40] (NCT0214996)	Phase 1, randomised, double-blind, placebo-controlled, multiple ascending dose	15–700 mg BID	Healthy volunteers, 36	2 weeks	Headache (13·9%); nausea (13·9%); AE ≥grade 3 (2%)	ATP ↑; 2,3-DPG decrease
Grace et al[58] (NCT02476916): DRIVE PK	Phase 2, open-label, single-arm	50 mg or 300 mg BID	PKD, not receiving regular RBC transfusions, 52	24 weeks (core); up to 8·5 years (extension)	Headache (44%); insomnia (40%); nausea (38%); AE ≥grade 3 (21%)	20 (38%) participants with Hb response ≥1·0 g/dL
Al-Samkari et al[59] (NCT03548220): ACTIVATE	Phase 3, randomised, placebo-controlled	Dose escalation from 5 mg to 50 mg BID	PKD, not receiving regular RBC transfusions, 40	24 weeks (core); 192 weeks (open-label extension)	Nausea (18%); headache (15%); AE ≥grade 3 (25%)	16 (40%) participants with sustained Hb response ≥1·5 g/dL
Glenthøj et al[60] (NCT03559699): ACTIVATE-T	Phase 3, open-label, single-arm	Dose escalation from 5 mg to 50 mg BID	PKD, receiving regular RBC transfusions, 27	40 weeks (core); 192 weeks (open-label extension)	ALT increase (37%); headache (37%); nausea (19%); AST increase (19%); fatigue (19%); AE ≥grade 3 (30%)	10 (37%) participants with transfusion reduction ≥33%
van Beers et al[61] (NCT03853798) ACTIVATE extension	Phase 3, open-label, single-arm	Dose escalation from 5 mg to 50 mg BID	PKD, not receiving regular RBC transfusions, 78	96 weeks	..	Improvements in laboratory markers of erythropoiesis and iron overload reduction in MRI liver iron content
Kuo et al[8] (NCT0369205)	Phase 2, open-label, single-arm	Dose escalation from 50 mg to 100 mg BID	α or β-NTDT, 20	24 weeks (core); 10 years (extension)	Insomnia (50%); dizziness (30%); headache (25%); AE ≥grade 3 (25%)	16 (80%) participants with Hb response of ≥1·0 g/dL
Taher et al[62] (NCT04770753): ENERGIZE	Phase 3, double-blind, randomised, placebo-controlled	100 mg BID	α or β-NTDT, 129	24 weeks (core); 5 years (open-label extension)	Headache (22%); insomnia (15%); nausea (12%); AE ≥grade 3 (14%)	55 (42%) participants with Hb response of ≥1·0 g/dL *vs* 1 (2%) participant in the placebo group
Xu et al[63] (NCT0400016)	Phase 1, open-label, single-arm	Dose escalation from 5 mg to 100 mg BID	SCD HbSS, 17	2 weeks each dose (core); up to 6 years (extension)	Insomnia (41%); hyperglycaemia (29%); pain (29%); AE ≥grade 3 (41%)	9 (56·3%) participants with Hb response ≥1·0 g/dL
Conrey et al[64] (NCT04000165) and Xu et al[63] extension	Phase 2, open-label, single-arm	Dose escalation from 50 mg to 100 mg BID	SCD HbSS, 15	Up to 144 weeks	VOCs (66·7%); estradiol ↓ (53·3%); esterone ↓ (46·7%); AE ≥grade 3 (73·3%)	14 (93%) participants with Hb response ≥1·0 g/dL
van Dijk et al[65] (EudraCT 2019-003438-18): ESTIMATE	Phase 2, open-label, single-arm	Dose escalation from 20 mg to 100 mg BID	SCD (HbSS, HbS-β^0 thal, or HbS-β^+ thal), 10	8 weeks	Headache (44%); ALT ↑ (44%); AST ↑ (22%); dyspepsia (22%); AE ≥grade 3 (not specified)	Mean reduction in point of sickling of 9·7 ±6·2 mmHg (p=0·0032) from baseline
van Dijk et al[7] (EudraCT 2019-003438-18): ESTIMATE extension	Phase 2, open-label, single-arm	Dose escalation from 20 mg to 100 mg BID	SCD (HbSS, HbS-β^0 thal, or HbS-β^+ thal), 9	52 weeks (core); 2 + 2 years (extension)[66]	ALT ↑ (70%); AST ↑ (60%); headache (40%); AE ≥grade 3 (20%)	Mean reduction in point of sickling of 4·1 ± 5·6 mm Hg (p=0·0802) from baseline
Idowu et al[67] (NCT05031780): RISE UP	Phase 2/3, double-blind, randomised, placebo-controlled	50 mg or 100 mg BID	SCD (any genotype), 52	12 weeks (phase 2); 52 weeks (phase 3); 216 weeks (open-label extension)	Headache (23%); arthralgia (15%); AE ≥grade 3 (15%)	12 (46%) participants with Hb response ≥1·0 g/dL in the 50 mg group and 13 (50%) in the 100 mg group *vs* one (4%) in the placebo group after 12 weeks of treatment

(Table 2 continues on next page)

	Design	Dosage	Population, n	Treatment period	Safety outcomes*	Key efficacy outcomes
(Continued from previous page)						
Etavopivat						
Forsyth et al[42] (NCT0381569)	Phase 1, randomised, placebo-controlled, double-blind, single ascending dose	200–1000 mg, single dose	Healthy volunteers, 24	Single dose	Each reported event occurred once; AE ≥grade 3 (4%)	··
Forsyth et al[42] (NCT0381569)	Phase 1, randomised, placebo-controlled, double-blind, multiple ascending dose	100, 200, 300 BID or 400 mg QD	Healthy volunteers, 36	2 weeks	Headache (28%); all other events occurred once; AE ≥grade 3 (0%)	ATP ↑; 2,3-DPG decrease; p50 decrease
Saraf et al[68] (NCT0381569)	Phase 1, randomised, placebo-controlled, single dose	700 mg, single dose	SCD (HbSS, HbS-β^0 thal, HbS-β^+ thal, or HbSC), 5	Single dose	Each reported event occurred once; AE ≥grade 3 (0%)	··
Saraf et al[68] (NCT0381569)	Phase 1, randomised, placebo-controlled, double-blind, multiple ascending dose	300 mg or 600 mg QD	SCD (HbSS, HbS-β^0 thal, HbS-β^+ thal, HbSC), 16	2 weeks	Headache (37·5%); nausea (37·5%); dizziness (25%); AE ≥grade 3 (12·5%)	11 (73%) participants with a Hb response ≥1·0 g/dL; p50 decrease
Saraf et al[68] (NCT0381569)	Phase 1, open-label, single-arm	400 mg QD	SCD (HbSS, HbS-β^0 thal, HbS-β^+ thal, or HbSC), 15	12 weeks	Headache (46·7%); nausea (26·7%); dizziness (20%); upper respiratory infection (20%); AE ≥grade 3 (40%)	11 (73%) participants with a Hb response ≥1·0 g/dL; p50 decrease

Safety and efficacy results correspond to the core period specified for each study. Results from the extension periods are listed where available. n=number of patients treated with pyruvate kinase activator (placebo not included). AE=adverse event. 2,3-DPG=2,3-diphosphoglycerate. BID=twice daily. PKD=pyruvate kinase deficiency. RBC=red blood cell. Hb=haemoglobin. ALT=alanine aminotransferase. AST=aspartate aminotransferase. NTDT=non-transfusion dependent thalassaemia. SCD=sickle cell disease. VOC=vaso-occlusive crisis. Thal=thalassaemia. QD=once daily. *Percentages represent the proportion of patients in whom the corresponding events occurred. Only safety outcomes from the intervention arm are displayed in the case of placebo-controlled trials. For each study, the three most frequent reported events (occurring in ≥10% of participants) are presented.

Table 2: **Clinical trials with completed and published results**

improved antioxidant status were seen in mitapivat-treated mice with sickle cell disease, showing reduced erythrocyte reactive oxygen species.[53]

PK activators also improve the red blood cell hydration state. Ex vivo treatment with both mitapivat and tebapivat significantly improved red blood cell hydration in hereditary spherocytosis red blood cells and in some red blood cells affected by hereditary xerocytosis.[52,54] These improvements in hydration, along with improvements in red blood cell deformability, were also observed in sickled cells following mitapivat treatment.[55] Additionally, PK activation in ex vivo red blood cells affected by sickle cell disease resulted in reduced intracellular Ca^{2+} levels and enhanced Ca^{2+} efflux.[49,56]

PK activators support cytoskeletal integrity, leading to improved red blood cell deformability. Both mitapivat and tebapivat decreased the tyrosine phosphorylation of band 3 in a dose-dependent manner, likely through increased protein phosphatase activity. Additionally, mitapivat treatment increased membrane ankyrin-1 levels and ankyrin-1 and band 3 interactions.[56]

PK activators might ameliorate ineffective erythropoiesis. Mitapivat enhances erythropoiesis and erythroid maturation and decreases intramedullary apoptosis in mice with β-thalassemia.[44] In mice with β-thalassemia and sickle cell disease, mitapivat also improves mitochondrial dysfunction during erythropoiesis by stimulating mitochondrial biogenesis and clearance.[44,53] In addition, mitapivat reduces reactive oxygen species within the erythroblasts and improves both erythroblast ATP production and resistance to oxidative stress.[44,57]

PK activation across hereditary haemolytic anaemias: evidence from clinical trials

PK activators have been evaluated in multiple clinical trials across a wide range of haematological disorders (tables 2 and 3).

Efficacy outcomes

PK deficiency

PK activators target the enzyme deficiency causing haemolytic anaemia. A phase 2 open-label clinical trial of mitapivat in adults with PK deficiency who were not receiving regular transfusions was initiated in 2015.[58] A rapid and sustained haemoglobin response of 1·0 g/dL was reached by 50% of the participants. These improvements were accompanied by a decrease in haemolytic markers and reticulocyte count. A relationship was found between PK protein levels at baseline, *PKLR* genotype, and haemoglobin response rate, with only those with at least one missense *PKLR* mutation achieving haemoglobin response.[58] The two phase 3 mitapivat trials in adults with PK deficiency therefore excluded patients with two non-missense mutations. The ACTIVATE trial was a phase 3, randomised, placebo-controlled trial in adults with PK deficiency who were not receiving regular transfusions, in which an early and sustained haemoglobin response of at least 1·5 g/dL was observed in 16 (40%) of 40 participants, compared with 0 of 40 participants receiving placebo.[59] The

ACTIVATE-T trial was an open-label, single-arm study conducted in adults with PK deficiency who were regularly transfused. Following mitapivat treatment, ten (37%) of 27 of the participants reached the transfusion reduction endpoint, with six (22%) reaching a transfusion-free status and three (11%) reaching a normal haemoglobin.[60] In addition, both trials showed improvements in haemolytic markers, reticulocyte count, and patient-reported outcomes, highlighting benefits on health-related quality of life.[69] In an extension study, mitapivat treatment also led to improvements in markers of iron overload, liver iron content, and erythropoiesis.[61] Based on these results, mitapivat is approved in multiple global regions for the treatment of symptomatic anaemia in adults with PK deficiency.

Following these results, two phase 3 randomised, placebo-controlled clinical trials of treatment with mitapivat in children with PK deficiency were initiated: ACTIVATE-KidsT (regularly transfused) and ACTIVATE-Kids (not regularly transfused).[70,71] Although neither study has yet been published, no new safety concerns in children have been reported to date.

	Design	Dosage	Population	Treatment period	Key outcome measures
Mitapivat					
NCT05175105: ACTIVATE-Kids	Phase 3, double-blind, placebo-controlled	Dose escalation from 1 mg to 50 mg BID	Paediatric (age <18 years) patients with non-transfusion-dependent PKD	20 weeks (core); 5 years (extension)	Hb response; sex hormones and BMD; haemolytic and erythropoietic markers; iron markers; patient-reported outcomes; pharmacokinetics and pharmacodynamics
NCT05144256: ACTIVATE-KidsT	Phase 3, double-blind, placebo-controlled	Dose escalation from 1 mg to 50 mg BID	Paediatric (age <18 years) patients with transfusion-dependent PKD	32 weeks (core); 5 years (open-label extension)	Transfusion reduction response; Hb response; sex hormones and BMD; iron markers; patient-reported outcomes; pharmacokinetics and pharmacodynamics
NCT04770779: ENERGIZE-T	Phase 3, double-blind, randomised, placebo-controlled	100 mg BID	Transfusion dependent α or β-thal	48 weeks (core); 5 years (open-label extension)	Transfusion reduction response; iron markers; safety; pharmacokinetics and pharmacodynamics
NCT06286046: RESIST	Phase 2, open-label, single-arm	100 mg BID	SCD (HbSS or HbS-β⁰ thal)	6 months	Kidney function and markers; hospitalisation rate; safety
NCT05935202: SATISFY	Phase 2, open-label, single-arm	Dose escalation from 50 mg to 100 mg BID	RBC membranopathies or CDA II	32 weeks (core); 1 year (extension)	Safety; Hb response; haemolytic and erythropoietic markers; iron markers; patient-reported outcomes
Etavopivat					
NCT04987489: GLADIOLUS	Phase 2, open-label	400 mg QD	SCD (cohort A); transfusion-dependent thal (cohort B); non-transfusion dependent thal (cohort C)	48 weeks	Hb response; transfusion reduction response; iron markers and overload
NCT06609226: FLORAL	Phase 3, open-label, roll-over	400 mg QD	SCD or thal who completed at least one treatment period in the parent study	264 weeks	Annualised rate of VOCs; Hb response; transfusion reduction response; hospitalisation rate; safety
NCT04624659: HIBISCUS	Phase 2/3, randomised, double-blind, placebo-controlled	200 mg or 400 mg QD	SCD (any genotype)	52 weeks (core); 112 weeks (open-label extension)	Hb response; annualised rate of VOCs; haemolytic and erythropoietic markers; patient-reported outcomes
NCT06612268: HIBISCUS 2	Phase 3, randomised, double-blind, placebo-controlled	400 mg QD	Patients aged ≥12 years with SCD (HbSS, HbS-β⁰ thal, or other syndrome)	52 weeks	VOC rate; Hb response; haemolytic and erythropoietic markers; patient-reported outcomes; kidney function
NCT05953584: HIBISCUS 3	Phase 2, open-label, single-arm	400 mg QD	Children with SCD (HbSS or HbS-β⁰ thal) who are at increased risk of stroke	52 weeks (core); 48 weeks (extension)	Timed average mean maximum velocity of the internal carotid artery and middle cerebral artery
NCT06198712: HIBISCUS KIDS	Phase 1/2, open-label, single-arm	400 mg QD	Paediatric (age 12–18 years) patients with SCD (any genotype)	24 weeks (core); 72 weeks (extension)	Pharmacokinetics and pharmacodynamics; Hb response; annualised rate of VOCs; patient-reported outcomes; timed average mean maximum velocity; safety
NCT05725902	Phase 2, open-label, single-arm	400 mg QD	Children with SCD (HbSS or HbS-β⁰ thal)	24 weeks	Cerebral blood flow; oxygen ejection fraction; cerebral metabolic rate of oxygen; safety
NCT05568225: FORTITUDE	Phase 2, open-label	400 mg QD	Very low, low, and intermediate risk MDS	48 weeks	Transfusion reduction response; Hb response; iron markers; platelet and neutrophil response; overall survival; safety

(Table 3 continues on next page)

	Design	Dosage	Population	Treatment period	Key outcome measures
(Continued from previous page)					
Tebapivat					
NCT04536792	Phase 1, randomised, double-blind, placebo-controlled, SAD and MAD	1 mg to 100 mg (SAD); 1 mg to 20 mg QD (MAD)	Healthy volunteers	Single dose (SAD); 2 weeks (MAD)	Safety; pharmacokinetics and pharmacodynamics
NCT04536792	Phase 1, open-label	2 mg or 5 mg QD	SCD (HbSS or HbS-β^0 thal)	4 weeks (core); 2 years (extension)	Safety; pharmacokinetics and pharmacodynamics; Hb response; haemolytic and erythropoietic markers
NCT06924970	Phase 2, double-blind, randomised, placebo-controlled	2·5 mg, 5 mg, or 7 mg QD	SCD (HbSS, HbSC, HbS-β^0 thal or other syndrome)	12 weeks (core); 52 weeks (extension)	Hb response; haemolytic and erythropoietic markers; patient-reported outcomes; safety
NCT05490446	Phase 2, open-label, single-arm (phase 2A)	5 mg QD	Lower-risk MDS*	16 weeks (core); 156 weeks (extension)	Hb response; transfusion independence; transfusion reduction response; safety; pharmacokinetics and pharmacodynamics
NCT05490446	Phase 2, open-label (phase 2B)	10 mg, 15 mg, or 20 mg QD	Lower-risk MDS*	24 weeks (core); 156 weeks (extension)	Transfusion reduction response; Hb response; safety; pharmacokinetics and pharmacodynamics

Safety measured as number and severity of treatment-emergent adverse events. BID=twice daily. PKD=pyruvate kinase deficiency. Hb=haemoglobin. BMD=bone mineral density. Thal=thalassaemia. SCD=sickle cell disease. RBC=red blood cell. CDA II=congenital dyserythropoietic anaemia type 2. QD=once daily. VOC=vaso-occlusive crisis. SAD=single ascending dose. MAD=multiple ascending dose. MDS=myelodysplastic syndrome. *Lower-risk MDS defined as Revised International Prognostic Scoring System risk score ≤3·5 and <5% blasts.

Table 3: **Ongoing clinical trials without completed and published core period results**

Thalassaemia

Thalassaemia is an inherited haemolytic anaemia characterised by impaired globin chain synthesis, causing imbalance between α-globin and β-globin chains in red blood cells. The precipitation of excess unpaired globin chains leads to oxidative stress and subsequent ineffective erythropoiesis, haemolysis, and anaemia.[34] To clear excess unpaired globin chains, thalassaemic red blood cells require more ATP but, paradoxically, have lower ATP levels.[72,73] Oxidative stress in thalassaemia is thought to inhibit PK activity by oxidising cysteine residues of PK.[74,75] In thalassaemia, PK activators address both ineffective erythropoiesis and haemolysis. An open-label phase 2 clinical trial in α and β non-transfusion dependent thalassaemia (NTDT) evaluated the safety and efficacy of mitapivat.[8] 16 (80%) of 20 participants across a wide spectrum of genotypes reached the primary endpoint, a haemoglobin increase of at least 1·0 g/dL. Two randomised phase 3 trials, ENERGIZE and ENERGIZE-T, were subsequently initiated to evaluate efficacy in NTDT and transfusion-dependent thalassaemia, respectively.[76,62] In ENERGIZE, haemoglobin responses of at least 1·0 g/dL were observed in 55 (42%) of 130 patients treated with mitapivat versus 1 (2%) of 64 patients treated with placebo.[62] These improvements were observed in both α-NTDT and β-NTDT, addressing a substantial unmet therapeutic need, especially in α-thalassaemia, for which no approved treatments are currently available. Although the ENERGIZE-T study results have yet to be published, preliminary findings indicate that mitapivat might decrease transfusion burden.[76] In both phase 2 and 3 trials in thalassaemia, mitapivat improved haemolytic, erythropoietic, and patient-reported outcomes, while iron markers were largely unchanged or unreported, underscoring the potential clinical benefit of PK activators in thalassaemia.

Sickle cell disease

Sickle cell disease results from a single nucleotide point mutation in the β-globin gene that produces mutated sickle haemoglobin, which polymerises upon deoxygenation.[1] Polymerisation of HbS distorts red blood cells to the characteristic sickle shape, increasing their fragility and causing haemolytic anaemia, while promoting microvascular obstruction that underlies painful vaso-occlusive events. Red blood cells in sickle cell disease are marked by reduced ATP levels, increased 2,3-DPG levels, with reduced oxygen affinity causing early deoxygenation.[6,15,77] The observed glycolytic disturbances in sickle cell disease might result from PK dysfunction.[6] These metabolic changes are associated with increased risk of polymerisation and poor outcomes.[15,78,79] Increased sickling and sickle cell disease-like phenotypes have been reported in individuals with sickle cell trait and co-inherited PK deficiency, supporting the hypothesis that PK activation could ameliorate sickling in this disease.[80,81] PK activators exert a dual effect in sickle cell disease by replenishing ATP and increasing haemoglobin–oxygen affinity. Mitapivat was evaluated in a phase 1, single-arm, open-label dose-escalation study, in which 9 of 16 participants had a haemoglobin response of at least 1·0 g/dL, along with reductions in haemolytic markers and reticulocyte count.[63] The investigator-initiated, single-arm phase 2 ESTIMATE study evaluated mitapivat in ten adults with sickle cell disease.[65] The primary endpoint, change in point of sickling, decreased after 8 weeks. Decreases in 2,3-DPG and increased haemoglobin–oxygen affinity shifted the oxygen dissociation curve towards normal. A haemoglobin response occurred in five of nine evaluable participants, with parallel reductions in reticulocytes and haemolytic markers. In subsequent extensions of mentioned studies with a median follow-up of 132 weeks and 156 weeks, these improvements were sustained.[64,66] Although exploratory and based on a small sample, the annualised rate of vaso-occlusive events in the ESTIMATE declined. This endpoint is being assessed in ongoing phase 3 trials. RISE UP was a phase 2/3, double-blind, randomised, placebo-controlled trial of mitapivat in adults with sickle cell disease. In the phase 2 part of this study, the primary efficacy endpoint, a haemoglobin response of at least 1·0 g/dL, was reached by 12 (46%) of 26 participants receiving 50 mg mitapivat twice daily and 13 (50%) of 26 receiving 100 mg mitapivat twice daily, compared with 1 (4%) of 27 receiving placebo. Haemolytic markers, reticulocytes, erythropoietin, and health-related quality of life also improved in both dosage groups. Although not powered for vaso-occlusive crises, annualised pain crises were 52–70% lower with mitapivat than with placebo.[67] These findings support continued evaluation of 100 mg twice daily in the phase 3 trial.

Etavopivat has also been evaluated in sickle cell disease. In a randomised, placebo-controlled, phase 1 study, a haemoglobin response of at least 1·0 g/dL was reached in 11 of 15 participants receiving etavopivat, along with improvements in ATP, 2,3-DPG, haemolytic markers, p50, point of sickling, and cellular deformability.[68] The ongoing phase 2/3 HIBISCUS trial is evaluating etavopivat in adults with sickle cell disease in a randomised, placebo-controlled, double-blind setting.[82] Although data have yet to be published, initial findings indicate a possible reduction in vaso-occlusive event rates.

Tebapivat has been evaluated in a phase 1, open-label study in adults with sickle cell disease. The complete trial results have not been published yet, but early findings suggest improvements in haemoglobin and haemolytic markers in both dosage groups (2 mg and 5 mg).[83]

Collectively, PK activators in sickle cell disease show improvements in haemoglobin, haemolytic and erythropoietic markers, health-related quality of life, and vaso-occlusive event rate, suggesting a potential clinical benefit in this patient population.

Red blood cell membrane disorders

Red blood cell membrane disorders arise from genetic defects in cytoskeletal, membrane and transmembrane, and ion-channel proteins and are characterised by haemolysis due to reduced deformability and subsequent splenic entrapment.[84] Glycolytic impairment is also observed, with reduced PK activity in hereditary spherocytosis, hereditary elliptocytosis, and hereditary xerocytosis.[4,5] In hereditary spherocytosis, PK deficiency likely reflects loss of membrane-associated glycolytic proteins as a result of membrane vesiculation.[4] Although PK activation does not correct the membrane defect, it improves red-cell rheology ex vivo irrespective of splenectomy status, supporting ongoing clinical trials.[52] Following these observations, the investigator-initiated phase 2 SATISFY trial evaluated mitapivat in hereditary spherocytosis, hereditary xerocytosis, and congenital dyserythropoietic anaemia type II.[85] Although results are not yet published, early data suggest improved haemoglobin, haemolytic markers, and patient-reported outcomes, particularly in hereditary spherocytosis.

Myelodysplastic syndrome

Myelodysplastic syndrome is an acquired disorder characterised by variable cytopenias, including anaemia due to ineffective erythropoiesis, and an increased risk of progression to acute myeloid leukaemia. Myelodysplastic syndrome has been associated with impaired glycolytic activity in red blood cells, which could be improved upon ex vivo treatment with PK activators.[86] A phase 2a/2b clinical trial investigating the safety and efficacy of tebapivat in lower-risk myelodysplastic syndrome (Revised International Prognostic Scoring System score ≤3·5 and <5% blasts) is ongoing.[87] Etavopivat has also been investigated for the treatment of myelodysplastic syndrome in an open-label phase 2 study, but the trial was terminated after enrolment of 17 out of 45 planned patients, as a futility assessment indicated a low likelihood of clinical benefit.[88]

Safety and tolerability

During the decade in which PK activators, mainly mitapivat, have been evaluated in humans, some potential safety risks have been identified. However, available data suggest that PK activators are generally well tolerated, and the reported adverse events are mostly mild, transient, and manageable with dose modifications or supportive interventions. The reported discontinuation rates in clinical trials across all diseases are low. The most common reported treatment-emergent adverse events are headache, insomnia, and nausea. Other frequently reported treatment-emergent adverse events included liver enzyme increase, dizziness, fatigue, sex hormone changes, and upper respiratory infections, while vaso-occlusive crises were reported in sickle cell disease (table 2).

Preclinical and early-phase clinical trials showed that mitapivat is an inhibitor of the aromatase enzyme. This is important because aromatase inhibition might cause hormone changes and reduce bone mineral density, and osteopenia and osteoporosis are more prevalent in patients with haemolytic disorders.[89] Adult trials have reported changes in sex hormones, albeit in most cases, not clinically significant.[58,64] These changes were reversible and did not lead to a significant decrease in bone mineral density, even after long-term treatment with mitapivat.[89,90]

Two patients with PK deficiency developed fractures after mitapivat initiation, one with prior L4 compression fracture and long-term steroid use.[59,91] Four traumatic single-digit fractures were reported in patients with sickle cell disease, mostly early after treatment start.[64] These findings emphasise the importance of continued assessment of bone mineral density.[59,91] Aromatase inhibition is also an important consideration in treatment of children, given potential effects on growth and pubertal development. This is being carefully examined in clinical trials of children with PK deficiency treated with mitapivat.[70,71] A phase 1 clinical ascending-dose study evaluating etavopivat found no aromatase inhibition.[42]

In the early phase of the DRIVE-PK study, two patients developed acute haemolysis after abrupt discontinuation of mitapivat.[58] After this occurrence, dose taper regimens were implemented in this trial and subsequent clinical trials. No events of acute haemolysis have been reported with dose decreases or discontinuing the medication with the implementation of these regimens.

Hepatocellular injury was reported in patients with thalassaemia treated with mitapivat. This occurred within the first 6 months of exposure. After treatment discontinuation, liver enzymes normalised. Hepatocellular injury was therefore included as a potentially important risk, and prescribing information was updated to include monthly liver enzyme monitoring during the first 6 months of treatment.[92]

The DRIVE-PK and ACTIVATE studies reported grade 3 or higher events of hypertriglyceridaemia in three (6%) of 52 and in two (5%) of 40 participants, respectively.[58,59] The ACTIVATE-T study also reported two occurrences, one of which was considered a grade 4 serious adverse event.[60] Screening for triglyceride increases should therefore be considered, and the long-term risk of cardiovascular disease should be weighed against the benefits of mitapivat treatment in those patients with hypertriglyceridaemia.

Concerns have been raised that drugs increasing haemoglobin–oxygen affinity could impair tissue oxygen delivery.[93,94] In sickle cell disease, voxelotor markedly limits oxygen release by stabilising the non-polymerising form of HbS. In contrast, PK activators lower 2,3-DPG, causing only a mild left shift of the oxygen–dissociation curve, while maintaining oxygen release and exerting anti-sickling effects.[95] Moreover, fetal haemoglobin (HbF) is less sensitive to 2,3-DPG mediated modulation of oxygen affinity than adult haemoglobin, as γ-globin binds 2,3-DPG more weakly, which is expected to attenuate the effect of 2,3-DPG changes in patients with high HbF proportions.[96] In the RISE UP study, this shift did not seem substantial enough to impair tissue oxygen delivery, as indicated by the absence of increased erythropoietin concentrations in either mitapivat group.[67] Neurological safety signals have not been reported to date, and an ongoing study (NCT05725902) is evaluating cerebral oxygen metabolism in patients with sickle cell disease receiving etavopivat.

Despite several identified potential risks, the available data support an acceptable safety profile for PK activators in the studied populations. Ongoing data collection will clarify which side-effects are class related. Given the potential for lifelong use, long-term monitoring and individualised risk-benefit assessment remain essential. Importantly, there are currently little long-term safety data.

Future directions

Many other red blood cell enzyme disorders could benefit from PK activation due to enhancement of glycolytic flux and restoration of metabolic and antioxidative homeostasis. A case report of a patient with phosphofructokinase deficiency described symptom improvement of muscle disease and improvements in haemoglobin following mitapivat treatment, supporting further exploration in other red cell enzymopathies, such as glucose-6-phosphate isomerase deficiency.[97] Ongoing trials of PK activators in haematological conditions, such as myelodysplastic syndrome, are expected to provide important insights that might support expanding their use to other hypoproliferative anaemias or bone marrow failure syndromes, such as Diamond-Blackfan anaemia.[87]

Ongoing evaluation in the paediatric population is important, as early initiation of treatment might prevent the development of long-term complications associated with haemolysis, blood transfusions, and iron overload. Most importantly, effective symptom management is crucial given the substantial disease burden during childhood, and this could prevent unnecessary splenectomy and its associated complications and risks.

Although mitapivat shows efficacy across a wide range of *PKLR* genotypes in patients with PK deficiency, most do not respond or only modestly.[58–60] Non-responders are often patients with minimal residual enzyme. This underscores the need for alternative therapeutic strategies, including the development of next-generation PK activators with efficacy in patients with severely impaired PK function.

Next-generation PK activators with high PKR and PKL selectivity, such as etavopivat, might prevent potential off-target, isoenzyme-related effects.[38] However, off-target PKM2 activation resulting in side-effects has not been investigated or reported. As PKL is more than 95% homologous to PKR and selectivity for PKR over PKL is not tested, it remains unclear whether mitapivat's side-effects will also emerge with etavopivat. Whether the improved pharmacodynamic and pharmacokinetic properties of next-generation PK activators, such as tebapivat, can improve outcomes has yet to be investigated.[39]

The rarity of many red blood cell disorders limits the feasibility of studying PK activation in clinical trials, while a substantial unmet therapeutic need remains. Continued evaluation through real-world data and basket trials involving cohorts with multiple rare anaemias should be encouraged, as these could improve our understanding of disease biology, provide further insights into the potential of PK activation, and expand the possibilities for targeted treatments in these ultra-rare haematological disorders.

Conclusions

PK activators represent a novel, oral, and generally well tolerated therapeutic approach for hereditary haemolytic anaemias, offering a mechanism-based alternative to supportive care. Clinical trials have shown meaningful improvements in key disease markers, with potential benefits in reducing complications and improving quality of life. An expanding number of red blood cell disorders could benefit from PK activation, supported by preclinical research suggesting a role of glycolytic disturbances in their pathophysiology. Early results of PK activation in children with PK deficiency suggest a favourable safety profile, supporting the expansion to early treatment, which remains an area of substantial unmet need. Second-generation PK activators with improved pharmacokinetic properties could optimise treatment further, and ongoing research underscores the expanding role of PK activation and its potential applications in other haematological conditions. Collectively, these recent advances have the potential to transform the clinical management of patients with hereditary haemolytic anaemias, particularly in diseases with few effective oral options, including sickle cell disease and thalassaemia.

Contributors

TD reviewed the literature and wrote the first draft of the manuscript, including figures and tables. AG, RFG, and EJvB reviewed and edited the manuscript. All authors approved the final version.

Declaration of interests

AG has received consultancy fees and is an advisory board member for Agios, Disc Medicine, Novo Nordisk, Pharmacosmos, and Vertex Pharmaceuticals, and has received research support from Agios Pharmaceuticals, Bristol Myers Squibb, Novo Nordisk, and Sanofi. RFG is a consultant for Agios Pharmaceuticals, Sanofi, and Sobi, an advisory board member for Sanofi, and unpaid medical advisory board member of Platelet Disorder Support Association and Rare Anemia International Network, and has received research funding from Agios Pharmaceuticals, Novartis, and Sobi. EJvB is consultant for Agios Pharmaceuticals, has received travel support and payment to the institution for presenting from Agios Phamaceuticals, has received research funding from Agios Pharmaceuticals, Vertex Pharmaceuticals, and Novo Nordisk, and holds roles in the Sickle Cell Outcome Register and ERN-EuroBloodNet. TD declares no conflicts of interest. During the preparation of this work the authors used OpenAI's language model, GPT-5, to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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